

KIRAN SANTHOSH REDDY

Sr. Full Stack Developer | Node.js & React.js

LinkedIn: www.linkedin.com/in/kiran-santhosh-reddy | Phone: +1 (737)-421-6095 | kiran.santhosh.reddy@gmail.com | Seattle, WA

PROFESSIONAL SUMMARY

- 10+ years of experience in full-stack web application development using Node.js, React.js, and cloud-native AWS architectures across healthcare, fintech, e-commerce, media, SaaS, and enterprise domains.
- Designed and developed scalable **Node.js microservices and REST/GraphQL APIs** using Express.js, Fastify, NestJS, and Apollo Server, deployed on **AWS (ECS, EKS, Lambda, EC2)**, serving millions of concurrent users with high availability and fault tolerance.
- Built complex, high-performance **React.js single-page applications (SPAs)** using React 18, Redux Toolkit, React Query, Next.js, and TypeScript, delivering pixel-perfect UIs with responsive design, accessibility (WCAG 2.1), and Core Web Vitals optimization.
- Expert in **React.js ecosystem** including React Hooks, Context API, Redux, MobX, React Router v6, React Testing Library, Storybook, Material UI, Tailwind CSS, and Styled Components for enterprise-grade frontend development.
- Proficient in **Node.js event-driven and asynchronous architectures**, implementing WebSocket real-time features, message queue consumers (Kafka, SQS, RabbitMQ), streaming APIs, and worker threads for CPU-intensive background processing.
- Strong expertise in **AWS cloud services** including Lambda, API Gateway, S3, CloudFront, ECS, EKS, RDS, DynamoDB, ElastiCache, Cognito, SQS, SNS, EventBridge, CDK, and CloudFormation for full-stack cloud-native application development.
- Experienced with **CI/CD pipelines using Jenkins, GitHub Actions, and AWS CodePipeline**, containerizing applications with Docker and Kubernetes, and enforcing code quality with ESLint, Prettier, SonarQube, Jest, Cypress, and Playwright.
- Proficient in **relational and NoSQL databases** including PostgreSQL, MySQL, MongoDB, DynamoDB, and Redis, using ORMs such as Sequelize, TypeORM, Prisma, and Mongoose for data access layer design in Node.js applications.
- Built **real-time collaborative features** using WebSockets (Socket.io), Server-Sent Events, and AWS API Gateway WebSocket APIs, enabling live dashboards, chat systems, collaborative editors, and event-driven notification platforms.
- Strong background in Agile/Scrum methodologies, leading UI architecture decisions, code reviews, sprint planning, and mentoring teams of 5–10 frontend and full-stack developers across globally distributed projects.

TECHNICAL SKILLS

Frontend – React.js: React 16/17/18, Next.js 13/14, TypeScript, JavaScript (ES6+), Redux Toolkit, React Query (TanStack), MobX, React Router v6, React Hook Form, Formik, Zustand, Context API, React Hooks, Suspense, Concurrent Mode

UI Libraries & Styling: Material UI (MUI v5), Ant Design, Tailwind CSS, Styled Components, Emotion, Bootstrap 5, Chakra UI, Sass/SCSS, CSS Modules, Storybook, Figma integration

Backend – Node.js: Node.js 18/20, Express.js, NestJS, Fastify, Koa.js, Apollo Server (GraphQL), REST APIs, GraphQL, WebSockets (Socket.io), gRPC, Serverless Framework, Microservices

Cloud – AWS: Lambda, API Gateway, S3, CloudFront, ECS, EKS, Fargate, EC2, RDS (PostgreSQL/MySQL), DynamoDB, ElastiCache (Redis), Cognito, SQS, SNS, EventBridge, Step Functions, CDK, CloudFormation, Secrets Manager, WAF, CloudWatch, X-Ray

Databases & Caching: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, Firebase Firestore, Sequelize, TypeORM, Prisma, Mongoose, Knex.js

Real-time & Messaging: Socket.io, WebSockets, AWS API Gateway WebSocket, Apache Kafka, AWS SQS, AWS SNS, RabbitMQ, Server-Sent Events (SSE), AWS EventBridge

Testing & Quality: Jest, React Testing Library, Cypress, Playwright, Vitest, Supertest, Mocha, Chai, Sinon, ESLint, Prettier, SonarQube, Istanbul (Code Coverage)

DevOps & CI/CD: Docker, Kubernetes, Helm, Jenkins, GitHub Actions, AWS CodePipeline, AWS CodeBuild, Terraform, AWS CDK, Git, GitHub, GitLab, Bitbucket, Nx Monorepo, Turborepo

Auth & Security: OAuth2, JWT, OpenID Connect, AWS Cognito, Auth0, Passport.js, OWASP Top 10, HTTPS/TLS, CORS, Helmet.js, Rate Limiting, CSRF protection

Other Languages & Tools: TypeScript, Python, Bash, SQL, GraphQL SDL, Webpack, Vite, Babel, esbuild, Rollup, npm, yarn, pnpm

Monitoring & Observability: AWS CloudWatch, Datadog, New Relic, Splunk, Sentry, LogRocket, Prometheus, Grafana, AWS X-Ray, PagerDuty

Methodologies: Agile/Scrum, SAFe, Micro-frontends, BFF (Backend for Frontend), Domain-Driven Design, TDD, BDD, SOLID Principles

PROFESSIONAL EXPERIENCE

Microsoft Corporation (Client: Azure DevOps): November 2023 – Current (Location: Seattle, WA (Hybrid))

Role: Sr. Full Stack Developer

Responsibilities:

- Architected and developed **React 18 + TypeScript micro-frontend applications** for the Azure DevOps Boards and Pipelines UI, using Module Federation (Webpack 5) to compose 12 independently deployable feature modules, reducing time-to-production for new features from 3 weeks to 4 days.
- Built **Node.js NestJS backend services** on AWS EKS serving Azure DevOps extension marketplace APIs, implementing GraphQL federation with Apollo Router, rate limiting, circuit breaking via Resilience patterns, and distributed caching with Redis ElastiCache for 50K+ daily active developers.
- Developed **real-time pipeline execution dashboards** using React 18 Concurrent Mode, WebSockets (Socket.io), and AWS API Gateway WebSocket APIs, streaming live build logs, test results, and deployment progress updates with <200ms latency to 100K+ concurrent users.
- Implemented **Next.js 14 App Router SSR/ISR** for Azure DevOps public documentation and marketplace listing pages, achieving 98 Lighthouse performance scores, Core Web Vitals (LCP <1.5s, FID <50ms, CLS <0.05), and 40% improvement in organic search indexing.
- Built **AWS Lambda + API Gateway REST and WebSocket APIs** in Node.js for DevOps extension event webhooks, processing GitHub, Jira, and Slack integration events with SQS-backed retry queues, DLQ handling, and CloudWatch structured logging.
- Designed **React component library using Storybook and MUI v5**, publishing 80+ reusable components to npm private registry with automated visual regression testing via Chromatic, reducing UI inconsistencies across 15 product teams and cutting design-to-code time by 60%.
- Implemented **end-to-end testing with Playwright** across 400+ test scenarios covering critical DevOps user journeys (pipeline creation, board configuration, artifact publishing), integrated into GitHub Actions CI with parallel execution reducing test suite runtime from 45 to 8 minutes.
- Built **AWS CDK TypeScript infrastructure stacks** provisioning EKS clusters, RDS Aurora PostgreSQL, ElastiCache Redis, API Gateway, CloudFront distributions, and WAF rules for the Azure DevOps extension backend, with blue/green deployment automation via CodeDeploy.

- Integrated **AWS Cognito and Auth0** for developer identity management in the extension marketplace, implementing OAuth2 PKCE flows, refresh token rotation, MFA enforcement, and seamless SSO with Microsoft Entra ID for enterprise customers.
- Established **frontend performance monitoring** using Datadog RUM (Real User Monitoring), custom React performance marks, and Core Web Vitals tracking, building dashboards identifying p95 render bottlenecks and reducing frontend error rate by 72% across Azure DevOps UI.

Environment: *React 18, TypeScript, Next.js 14, Redux Toolkit, Module Federation, Node.js, NestJS, GraphQL, Apollo Server, Socket.io, AWS EKS, AWS Lambda, API Gateway, ElastiCache Redis, RDS Aurora, AWS CDK, AWS Cognito, Auth0, Playwright, Storybook, MUI v5, Datadog, GitHub Actions, Docker, Kubernetes*

Stripe Inc.: April 2021 – October 2023 (Location: San Francisco, CA (Remote))

Role: Sr. Full Stack Developer

Responsibilities:

- Built **React 17 + TypeScript Stripe Dashboard features** for the payment analytics, dispute management, and radar fraud review modules, using Redux Toolkit and React Query for state management, serving 3M+ merchants processing \$1T+ in annual payment volume.
- Developed **Node.js Express.js microservices** powering Stripe Dashboard APIs for financial reporting, payout scheduling, and merchant onboarding, deployed on AWS ECS Fargate with ALB-based routing, auto-scaling, and multi-region active-active configurations for 99.999% availability.
- Implemented **real-time payment event streaming** using Apache Kafka consumers in Node.js, processing charge, refund, dispute, and payout webhook events and pushing live updates to merchant dashboards via Server-Sent Events (SSE) with backpressure handling.
- Built **Next.js 13 server-side rendered** merchant-facing documentation site and API reference, implementing ISR (Incremental Static Regeneration) for 10,000+ API reference pages, reducing TTFB from 2.1s to 180ms and eliminating client-side hydration flicker.
- Designed **GraphQL API layer using Apollo Server** and Node.js aggregating Stripe payment, customer, subscription, and invoice data for the Dashboard BFF (Backend for Frontend), implementing DataLoader batching to eliminate N+1 queries and query complexity limits for DoS protection.
- Implemented **PCI-DSS compliant React form components** for card data capture using Stripe Elements and Payment Element, custom iframe sandboxing, CSP headers, and OWASP-compliant XSS/CSRF protections integrated with AWS WAF for merchant checkout flows.
- Built **AWS Lambda@Edge functions** in Node.js for CloudFront-based request routing, A/B testing traffic splitting, geo-based merchant redirects, and JWT authentication at the CDN edge layer, reducing origin server load by 45%.
- Developed **Cypress E2E test suites** covering 300+ merchant Dashboard workflows including payment creation, refund processing, dispute responses, and payout configuration, integrated into GitHub Actions with parallel Cypress Cloud execution achieving 95% critical path coverage.
- Built **Node.js CLI tools and internal developer platform APIs** for Stripe engineering teams, automating API version upgrade migrations, SDK code generation, and changelog management, reducing manual release overhead by 8 hours per Stripe API version bump.
- Implemented **Sentry and LogRocket session replay integration** in the React Dashboard for real-time error tracking and UX debugging, establishing frontend error budgets and SLOs with automated PagerDuty alerting when merchant-impacting error rates exceeded thresholds.

Environment: *React 17, TypeScript, Redux Toolkit, React Query, Next.js 13, Node.js, Express.js, GraphQL, Apollo Server, Apache Kafka, AWS ECS Fargate, AWS Lambda@Edge, CloudFront, AWS WAF, PostgreSQL, Redis, Cypress, GitHub Actions, Docker, Kubernetes, Sentry, LogRocket, Datadog*

Netflix Inc. (Contract): January 2019 – March 2021 (Location: Los Gatos, CA)

Role: Sr. Full Stack Developer

Responsibilities:

- Developed **React 16/17 + TypeScript components** for the Netflix Studio Content Management Platform (Cosmos), enabling post-production teams to manage 10,000+ original content assets, metadata, delivery schedules, and localization workflows across 190 countries.
- Built **Node.js Fastify microservices** for the content delivery orchestration platform, managing video encoding job queues, subtitle processing pipelines, and multi-CDN asset distribution workflows using Apache Kafka for event coordination and DynamoDB for job state management.
- Implemented **React real-time collaborative editing features** using Operational Transformation (OT) and WebSockets (Socket.io) for concurrent script and metadata editing, enabling 50+ post-production editors to collaborate on the same content asset simultaneously without conflict.
- Designed **GraphQL federation gateway using Apollo Federation v2** aggregating Node.js subgraph services for content metadata, rights management, localization, and scheduling, enabling Studio teams to compose complex queries across 20+ domain services with a unified schema.
- Built **Next.js SSR applications** for the Netflix Partner Portal, enabling studio production companies and distributors to submit content, track delivery status, and manage royalty reporting, with AWS Cognito-based federated identity for external partner authentication.
- Implemented **AWS Step Functions workflows** orchestrated by Node.js Lambda functions for automated content QC pipelines, coordinating video analysis, audio normalization, subtitle compliance checks, and multi-format packaging with automatic retry and human-in-the-loop approval flows.
- Developed **custom React data visualization components** using D3.js and Recharts for the Studio analytics dashboard, displaying content performance metrics, streaming engagement data, and production cost tracking across 200+ original productions.
- Built **Node.js BFF (Backend for Frontend) services** on AWS ECS for the Studio web and mobile platforms, aggregating data from 15 internal microservices, implementing response caching with Redis ElastiCache, and GraphQL persisted queries for mobile bandwidth optimization.
- Implemented **Jest and React Testing Library unit test suites** achieving 88% code coverage across Studio UI components, establishing snapshot testing, accessibility audits (axe-core), and visual regression testing with Chromatic integrated into the GitHub Actions CI pipeline.

Environment: *React 17, TypeScript, Next.js, Redux, Node.js, Fastify, GraphQL, Apollo Federation, Socket.io, D3.js, Apache Kafka, AWS ECS, AWS Lambda, AWS Step Functions, DynamoDB, ElastiCache, Redis, AWS Cognito, AWS CDK, Jest, React Testing Library, Chromatic, GitHub Actions, Docker*

Salesforce Inc.: July 2017 – December 2018 (Location: San Francisco, CA)

Role: Full Stack Developer

Responsibilities:

- Built **React 15/16 Lightning Design System (LDS) components** for Salesforce Sales Cloud and Service Cloud, developing custom Einstein Analytics dashboard widgets, opportunity management views, and case management flows consumed by 150,000+ enterprise CRM users.
- Developed **Node.js Express.js REST APIs** for Salesforce AppExchange integrations, building OAuth2 connected app flows, Apex REST callout handlers, and webhook processors for HubSpot, Zendesk, DocuSign, and Marketo third-party ecosystem connectors.
- Implemented **Redux and React Context state management** for complex multi-step CPQ (Configure, Price, Quote) workflows in the Salesforce Sales Cloud UI, managing product configuration trees, pricing rule evaluations, and approval chain states across 10+ wizard steps.
- Built **Node.js Salesforce data loader microservices** on AWS Lambda for automated bulk record operations, implementing the Salesforce Bulk API 2.0 for nightly CRM data sync, territory assignment updates, and lead scoring refreshes for enterprise accounts.

- Developed **real-time Salesforce event streaming consumers** using Node.js and Salesforce Platform Events (CometD protocol), processing opportunity stage changes, case escalations, and approval decisions to trigger downstream Slack, email, and SMS notification workflows via AWS SNS.
- Built **TypeScript utility libraries** for Salesforce SOQL query building, schema metadata caching, and field permission enforcement, published as internal npm packages used by 30+ Salesforce development teams to standardize data access patterns.
- Implemented **AWS API Gateway and Lambda-based Salesforce webhook receivers**, processing Salesforce outbound messages and platform events with SQS buffering, idempotent processing, and CloudWatch alerting for CRM integration reliability monitoring.

Environment: *React 16, TypeScript, Redux, Node.js, Express.js, Salesforce LWC, Apex, Salesforce Bulk API, AWS Lambda, API Gateway, AWS SQS, AWS SNS, PostgreSQL, MongoDB, Redis, Jest, Mocha, GitHub Actions, Docker, Jenkins, Webpack*

Uber Technologies Inc.: February 2015 – June 2017 (Location: San Francisco, CA)

Role: Full Stack Developer

Responsibilities:

- Developed **React.js + Flux/Redux dashboard applications** for Uber's internal Operations and Driver partner portals, building trip management, earnings analytics, incentive program management, and support case tracking tools used by 5,000+ operations staff globally.
- Built **Node.js Express microservices** for Uber's driver onboarding platform, implementing document verification workflows, background check integrations, vehicle registration processing, and onboarding status state machines serving 500K+ new driver applications monthly.
- Implemented **React Native web-compatible components** using React Native Web for Uber Driver app web views, enabling code sharing between native mobile and web platforms for earnings summaries, weekly statements, and tax document download features.
- Built **Node.js Kafka consumers** processing real-time trip completion events, fare calculation messages, and surge pricing updates, aggregating data into driver earnings summaries and pushing live notifications via WebSockets to the driver partner portal dashboard.
- Developed **AWS Lambda functions in Node.js** for serverless trip receipt generation, tax document creation (1099-K, 1099-NEC), and earnings export processing, triggered by SQS events with S3 output delivery and CloudFront-signed URL distribution to drivers.
- Implemented **React.js data visualization dashboards** using Chart.js and custom SVG components for Uber's City Operations teams, displaying real-time supply/demand heatmaps, driver utilization metrics, and market performance KPIs across 600+ global cities.
- Built **Node.js CLI tooling and internal developer APIs** for Uber engineering teams, automating microservice scaffolding, API contract testing, and deployment configuration generation, reducing new service bootstrap time from 2 days to 30 minutes.

Environment: *React.js, Redux, JavaScript (ES6), Node.js, Express.js, Apache Kafka, AWS Lambda, AWS SQS, AWS S3, CloudFront, MongoDB, Redis, PostgreSQL, WebSockets, Chart.js, Jest, Mocha, Jenkins, Docker, Webpack, Babel*

Infosys Limited (Client: Citibank NA): August 2013 – January 2015 (Location: New York, NY)

Role: Full Stack Developer

Responsibilities:

- Developed **React.js (early adoption) and AngularJS frontend applications** for Citibank's Retail Internet Banking portal, building account summary, fund transfer, bill payment, and statement download features serving 12M+ online banking customers with responsive design across desktop and mobile browsers.

- Built **Node.js Express REST APIs** for Citibank's banking middleware integration layer, consuming core banking SOAP/XML services and exposing clean JSON REST endpoints for web and mobile banking clients, implementing OAuth2 token management and session security.
- Implemented **real-time transaction notification features** using Node.js WebSockets and long-polling fallback, delivering instant debit/credit alerts, fraud alerts, and payment confirmation messages to online banking customers within 2 seconds of transaction posting.
- Built **jQuery and Bootstrap 3 responsive UI components** for Citibank's legacy banking portals, progressively migrating UI modules to React.js, implementing cross-browser compatibility fixes and WCAG 2.0 accessibility standards for screen reader support.
- Developed **Node.js batch processing scripts** for nightly statement generation, eStatement delivery, and transaction categorization, integrating with AWS S3 for encrypted PDF storage and SES for email delivery to 8M+ enrolled eStatement customers.
- Implemented **PCI-DSS compliant form validation and data masking** in React.js for credit card application flows, account number display masking, and CVV entry fields, integrating with Citibank's HSM-based encryption APIs for sensitive card data handling.

Environment: *React.js, AngularJS, JavaScript, jQuery, Bootstrap, Node.js, Express.js, REST APIs, SOAP/XML, WebSockets, AWS S3, AWS SES, PostgreSQL, Oracle, Redis, Jenkins, Git, Webpack, Babel, Mocha, Chai*

CERTIFICATIONS

AWS Certified Developer – Associate – Amazon Web Services (AWS) | Issued: February 2023 | Credential ID: AWS-DVA-2023-KR8812

AWS Certified Solutions Architect – Associate – Amazon Web Services (AWS) | Issued: August 2021 | Credential ID: AWS-SAA-2021-KR5534

AWS Certified Cloud Practitioner – Amazon Web Services (AWS) | Issued: March 2020 | Credential ID: AWS-CCP-2020-KR2201

Meta (Facebook) Certified React Developer – Meta / freeCodeCamp | Issued: September 2022 | Credential ID: META-RD-2022-KR4491

Node.js Application Developer Certification (JSNAD) – OpenJS Foundation | Issued: November 2021 | Credential ID: JSNAD-2021-KR-00341

Node.js Services Developer Certification (JSNSD) – OpenJS Foundation | Issued: April 2022 | Credential ID: JSNSD-2022-KR-00512

Certified Kubernetes Application Developer (CKAD) – Cloud Native Computing Foundation (CNCF) | Issued: January 2023 | Credential ID: CKAD-2023-KR-00889

GraphQL Associate Certification – Apollo GraphQL | Issued: July 2022 | Credential ID: APOLLO-GA-2022-KR3312

HashiCorp Certified: Terraform Associate – HashiCorp | Issued: October 2022 | Credential ID: HCTA-2022-KR-00678

MongoDB Certified Developer Associate – MongoDB University | Issued: May 2021 | Credential ID: MDB-DEV-2021-KR7723

Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft | Issued: June 2020 | Credential ID: AZ900-2020-MS-KR5521

Google Associate Cloud Engineer – Google Cloud (GCP) | Issued: December 2022 | Credential ID: GCP-ACE-2022-KR9901

EDUCATION

Bachelor of Engineering in Information Technology – Visvesvaraya Technological University (VTU),
Bangalore, India, 2009–2013